

QED-Net

A Hybrid Quantum–Classical Machine Learning Platform
for Early Disease Detection

Project Documentation and Technical Design

Version 1.0

Team Member A (Quantum Research Lead)
Team Member B (Data & Evaluation Lead)
Team Member C (Platform & Integration Lead)

Project Guide: _____
Institution: _____

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Research Use Only. This platform is a research and educational artifact. It is *not* a medical device, is *not* certified for clinical use, and must never be used to inform real patient care. All experimental results in this project are benchmark studies on public datasets.

Abstract

QED-Net (**Q**uantum-**E**nhanced **D**etection **N**etwork) is a hybrid quantum–classical machine learning platform for early disease detection on biomedical data. The system ingests clinical and biomedical datasets, applies leak-proof classical pre-processing and feature compression, and feeds the compressed representation into quantum models (quantum-kernel SVMs, variational quantum classifiers, data re-uploading circuits, and hybrid CNN–quantum networks) executed on high-performance simulators and, for validation, on real quantum hardware.

The project takes an explicitly *honest* scientific position: published evidence does not support the claim that quantum models outperform well-tuned classical models on real biomedical data. Instead of chasing accuracy headlines, QED-Net is built around three defensible contributions: (i) a rigorous, leakage-proof evaluation harness with statistical testing, calibration analysis and cost accounting; (ii) a *Quantum Advantage Certificate* — an automated referee that adversarially tests whether quantum components provide any measurable benefit, including a parameter-matched entanglement-ablation protocol and a classical kernel-approximation (dequantization) attack; and (iii) deployment-oriented features (a classical–quantum cascade, conformal prediction with abstention, and shot-noise-derived uncertainty) that make small, slow quantum models usable inside a realistic clinical workflow.

This document specifies the system architecture, terminology, technology stack, feature set, evaluation methodology, team organisation for a three-member team, and a complete contingency plan for solo execution.

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Chapter 1

Introduction

1.1 Background

Early and accurate disease detection improves treatment outcomes and reduces cost. Classical machine learning (especially gradient-boosted trees such as XGBoost) is the current state of the art on tabular biomedical data. Quantum machine learning (QML) proposes to use quantum circuits as learnable models, motivated by the high-dimensional Hilbert spaces in which quantum states live. Because present-day quantum computers are small and noisy, a *hybrid* architecture — classical pre-processing, a small quantum core, classical post-processing — is the only executable design today.

1.2 Problem Statement

Design and build a hybrid quantum–classical machine learning platform for early disease detection that supports: data ingestion, hybrid model training, prediction, explainability, and rigorous performance evaluation against strong classical baselines, on real or benchmark biomedical datasets.

1.3 Objectives (Revised and Honest)

The original objective “improve accuracy, sensitivity and specificity over classical baselines” is not scientifically defensible: no reproducible study shows quantum models beating *properly tuned* classical models on real biomedical data. QED-Net therefore reframes the objectives as follows:

1. **O1 — Architecture.** Deliver a working hybrid quantum–classical architecture for biomedical classification.
2. **O2 — Models.** Implement four quantum model families (QSVM, VQC, data re-uploading, hybrid QNN) that process high-dimensional data via a classical compression stage.
3. **O3 (revised) — Measured advantage, not assumed advantage.** Quantify where quantum helps using three honest axes: *parameter efficiency* (e.g., a 12-parameter circuit vs. a 400-parameter MLP), *small-data robustness* ($n = 50\text{--}200$), and *statistically tested performance parity* (within confidence intervals of XGBoost). Report all results, positive or negative.
4. **O4 — Rigor.** Produce a leak-proof, reproducible, multi-seed, statistically tested benchmark harness — arguably the most valuable deliverable, because no such harness exists for QML-vs-classical comparisons on medical data.
5. **O5 — Deployability.** Wrap models in an API, dashboard, calibration, uncertainty estimation, and abstention so that the system behaves like a research product rather than a notebook.

The three hard truths that shape this design.

1. ***The premise is marketing.*** *There is no reproducible quantum-over-classical result on real biomedical data; kernel advantages are provable only on artificially constructed problems.*
 2. ***The accuracy objective is a trap.*** *XGBoost reaches 97–98% AUROC on the standard benchmarks. A claim of beating it implies a weak baseline, data leakage, or a cherry-picked seed — all detectable by a competent judge.*
 3. ***The QML–medical literature is methodologically weak.*** *We copy its circuits and reject its evaluation practices.*
-

1.4 Scope

In scope: tabular and signal datasets (breast cancer, heart disease, Parkinson’s voice, PTB-XL ECG), four quantum model families, five classical baselines, full evaluation stack, explainability, cascade/conformal/uncertainty deployment features, one hardware run, API + dashboard, documentation.

Out of scope (documented as Future Work, Section 15): quantum survival analysis, mixed-state missing-data encoding, cross-modal entanglement fusion, federated training, clinical deployment.

Chapter 2

Terminology and Foundational Concepts

This chapter explains every term used in the project. Each entry is given twice: *plain words* (no background required) and *technically* (precise definition).

2.1 Quantum Computing Terms

Table 2.1: Quantum terminology.

Term	In plain words	Technically
Qubit	A coin spinning in the air: it is “both” heads and tails until it lands. Computers store answers in these coins.	A two-level quantum system; state $ \psi\rangle = \alpha 0\rangle + \beta 1\rangle$ with complex amplitudes, $ \alpha ^2 + \beta ^2 = 1$. Measurement yields 0 with probability $ \alpha ^2$.
Superposition	Being in many states at once. 20 qubits blend about a million possibilities simultaneously.	The state vector of n qubits lives in a 2^n -dimensional Hilbert space; the amplitudes encode a weighted blend over all basis states.
Entanglement	Two qubits behaving as one linked object; measuring one instantly constrains the other. The genuinely non-classical resource.	A state that cannot be factorised as a product of single-qubit states; quantified for a bipartition $A B$ by the entanglement entropy $S(\rho_A) = -\text{Tr}(\rho_A \log_2 \rho_A)$.
NISQ era	Today’s quantum computers: small, noisy prototypes.	<i>Noisy Intermediate-Scale Quantum</i> : tens–hundreds of physical qubits, gate fidelities $< 99.9\%$, no error correction.
Quantum circuit	A sequence of operations (gates) applied to qubits, like a recipe of rotations.	A product of unitary matrices $U = U_L \cdots U_2 U_1$ acting on the state vector; parameterised circuits $U(\theta; x)$ carry trainable angles θ and encoded data x .

Term	In plain words	Technically
Gate	One instruction: rotate one qubit, or link two.	Unitary operator; single-qubit rotations $R_X(\theta) = e^{-i\theta X/2}$ etc.; entangling gates (CNOT, CZ, R_{ZZ}) act on pairs and create entanglement.
Measurement (shot)	One look at the spinning coin; it lands and you record the result.	A projective measurement in a chosen basis; one execution of the circuit is a “shot”. Expectation values, e.g. $\langle Z \rangle$, are estimated from S shots.
Shot noise	The unavoidable wobble in results because you only looked a limited number of times.	Sampling variance of an estimated probability: $\text{Var}(\hat{p}) = p(1-p)/S \leq 1/(4S)$.
Statevector simulator	A normal computer pretending to be a quantum one by tracking all amplitudes.	Classical simulation storing all 2^n amplitudes; memory and time grow exponentially, capping practical simulation at ~ 25 qubits (CPU).
Density matrix ρ	A way to describe a qubit when you are <i>genuinely unsure</i> , not just blending on purpose.	Positive semi-definite trace-1 operator; mixed states represent classical uncertainty; $\rho = I/2$ is the maximally mixed state (carries zero information). Simulation cost $O(4^n)$.
Barren plateau	Training silently dies: gradients shrink to almost zero as circuits grow, so nothing learns.	For random deep circuits with global costs, $\text{Var}(\partial_\theta f)$ decays <i>exponentially</i> in qubit count and depth [2]. Mitigations: shallow circuits, local costs, layerwise training.
Encoding (feature map)	Translating patient numbers into rotations of qubits — the only language qubits understand.	A map $x \mapsto \varphi(x)\rangle$, e.g. angle encoding ($R_X(x_i)$ per feature) or ZZFeatureMap (Hadamards + R_{ZZ} layers) [1].
Variational circuit / ansatz	The trainable “brain”: a circuit with adjustable knobs tuned by a normal optimiser.	$U(\theta; x)$ with classical optimisation of a loss $\mathcal{L}(\theta)$; gradients via backprop-through-simulator, the parameter-shift rule $\frac{\partial f}{\partial \theta_k} = \frac{f(\theta_k + \pi/2) - f(\theta_k - \pi/2)}{2}$, or SPSA.
Transpilation	Rewriting your circuit into the version a real (awkwardly wired) quantum chip can actually run.	Decomposition into the hardware’s native gate set respecting its coupling map; typically increases depth and two-qubit gate count.

2.2 Machine Learning and Data Terms

Table 2.2: Machine learning and data terminology.

Term	In plain words	Technically
Features / labels	The measurements describing each patient / the known answer (diseased or not).	Design matrix $X \in \mathbb{R}^{n \times d}$, target vector $y \in \{0, 1\}^n$.
Supervised learning	Studying answered past exams so you can answer future ones.	Fitting $f_\theta : \mathcal{X} \rightarrow [0, 1]$ by minimising empirical risk on labelled pairs (x_i, y_i) .
Cross-validation (CV)	Rotating practice matches so every datapoint gets scored once on unseen turf.	k -fold partitioning; performance averaged over folds. <i>Nested</i> CV adds an inner loop for hyperparameter tuning so the outer score is unbiased.
Data leakage	Studying the real exam the night before — great marks, zero learning. The single most common fatal flaw.	Any information from the test fold influencing training: fitting scalers/imputers/SMOTE on the full dataset before splitting, feature selection on all data, target leakage via post-outcome variables.
SMOTE	Inventing synthetic minority-class patients so the classes are balanced — but only <i>inside</i> training folds, or it leaks.	Synthetic Minority Oversampling: interpolates minority samples; must be applied within CV folds only.
PCA / autoencoder	Squeezing 30–20,000 numbers into 8–16 summary numbers with minimal information loss.	PCA: linear projection to top-variance directions, $z = Wx$. Autoencoder: neural net trained to reconstruct its input through a bottleneck layer.
XGBoost	The current champion algorithm for table-shaped medical data; our real opponent.	Gradient-boosted decision trees [7]; tuned with Optuna (50+ trials) in this project.
Calibration (Platt / isotonic)	Making “70% risk” actually mean 70 out of 100.	Post-hoc mapping of scores to probabilities via a fitted sigmoid (Platt) or monotonic step function (isotonic) on a held-out set.
SHAP	An attribution method answering “which inputs pushed this prediction?” by fairly dividing the credit.	Shapley values from cooperative game theory; KernelSHAP approximates them for any black-box pipeline [8].

Term	In plain words	Technically
Conformal prediction	A wrapper that lets the model say “I don’t know” with a mathematical guarantee of how often it may be wrong.	Split-conformal sets $C(x)$ with finite-sample coverage $\Pr[y \in C(x)] \geq 1 - \alpha$ under exchangeability [9]; an uninformative set = abstain.
Model card	A nutrition label for a model: intended use, data, performance, known failure modes.	Structured documentation per Mitchell et al. [10]; mandatory for every trained model in QED-Net.

2.3 Evaluation and Clinical Terms

Table 2.3: Evaluation and clinical terminology.

Term	In plain words	Technically
AUROC (AUC)	“Given one sick and one healthy patient, how often does the model rank the sick one as riskier?” 0.5 = coin flip; 1.0 = perfect.	Probability that a random positive scores above a random negative; compared statistically between models with the DeLong test [11].
AUPRC	The honest metric when disease is rare; ignores the easy true negatives.	Area under precision–recall curve; baseline equals disease prevalence.
Sensitivity / specificity	Of the truly sick, how many are caught / of the truly healthy, how many are correctly cleared.	$TP/(TP + FN)$ and $TN/(TN + FP)$; <i>sensitivity at 95% specificity</i> is the clinically operative operating point.
MCC	A single balanced score that stays honest under class imbalance.	$MCC = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}$
Brier score / ECE	How trustworthy the probability numbers are.	$\text{Brier} = \frac{1}{N} \sum_i (p_i - y_i)^2;$ $\text{ECE} = \sum_b \frac{n_b}{N} \text{acc}_b - \text{conf}_b $ over confidence bins b .
Bootstrap confidence interval	Resampling the test set 2,000 times to get an honest error bar.	Percentile CI over bootstrap replicates of the metric.
Decision curve analysis (DCA)	“Would doctors using this model achieve more net benefit, counting the harm of false alarms?”	Net benefit = $\frac{TP}{n} - \frac{FP}{n} \cdot \frac{t}{1-t}$ at threshold probability t , vs. “treat all” and “treat none”.

Term	In plain words	Technically
Early detection	Finding disease <i>before</i> symptoms — fundamentally a time-to-event question.	Formally a longitudinal/survival problem; in this project we treat the cross-sectional surrogate, and document survival analysis as future work.
EHR / genomics	Electronic hospital records / measurements of gene activity (expression).	EHR: irregular, 30–60% missing, informative missingness. Genomics: $d \approx 20,000$ features for $n \approx 1,000$ samples (the small- n , large- d regime).

2.4 Notation

Symbol	Meaning
$x \in \mathbb{R}^d, y \in \{0, 1\}$	patient feature vector; label
n, d	number of samples; number of raw features
n_q	number of qubits (typically 8–16)
S	number of shots (circuit repetitions)
θ	trainable circuit parameters
$U(\theta; x)$	parameterised quantum circuit
$ \varphi(x)\rangle$	encoded data state
$K(x, y) = \langle \varphi(x) \varphi(y) \rangle ^2$	quantum kernel
$\langle Z \rangle$	Pauli- Z expectation value (model output pre-sigmoid)
$\rho = I/2$	maximally mixed (zero-information) state
$S(\rho_A)$	entanglement entropy of subsystem A

Table 2.4: Mathematical notation used throughout.

Chapter 3

System Architecture

3.1 The Pipeline

Figure 3.1 shows the eight-stage pipeline. The quantum core (stages 4–6, shaded) is deliberately narrow: current hardware and simulators restrict inputs to roughly 8–16 dimensions.

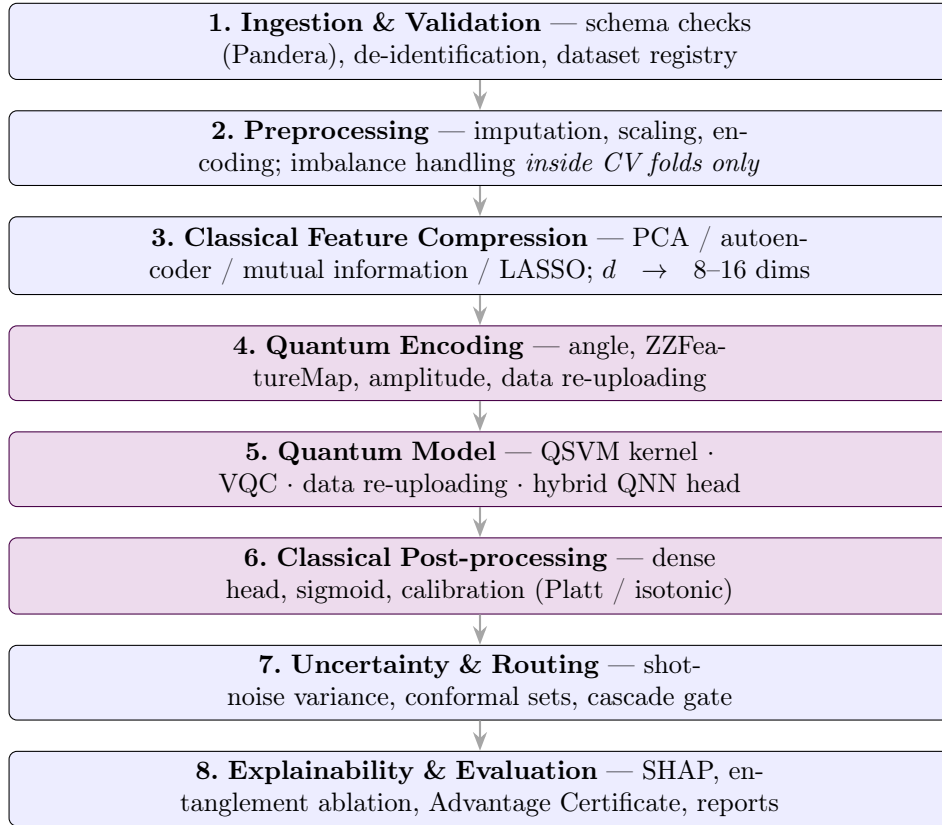


Figure 3.1: QED-Net end-to-end pipeline. The shaded stages form the quantum core; stage 3 (compression) performs the bulk of the representational work and this is stated openly in all documentation.

Design fact stated openly. *Current simulators and hardware cap the quantum core at 8–16 features. The classical compression stage therefore does ~95% of the representational work. QED-Net measures this explicitly (Bottleneck Honesty Meter, Section 9.2) rather than hiding it. Teams that hide it are dismantled in Q&A; teams that measure it are trusted.*

3.2 Module Map

The platform is organised as one Python package with clear sub-packages, mirrored by the deliverables D1–D20 (Section 9.6):

- `qednet.data` — ingestion, validation, de-identification (D2)
- `qednet.preprocess` — imputation, scaling, selection, compression (D3)
- `qednet.encoding` — quantum feature maps (D4)
- `qednet.models.quantum` — QSVM, VQC, re-uploading, hybrid QNN (D5)
- `qednet.models.classical` — baselines (D6)
- `qednet.train` — config-driven training, checkpoints (D7)
- `qednet.evaluation` — nested CV, metrics, statistics, certificate (D8)
- `qednet.explain` — SHAP, ablation, sensitivity (D9)
- `qednet.deploy` — cascade, conformal, uncertainty (F1/F6/N5)
- `api/`, `dashboard/` — FastAPI service, web UI (D11, D12)

3.3 Interface Contract (critical for teamwork)

Every model — quantum or classical — implements the scikit-learn-style interface `fit(X, y)` / `predict_proba(X)`. Every pipeline step is a standard transformer with `fit` / `transform`. This single contract is what allows three people to build models, evaluation, and platform in parallel without blocking each other, and it is what the leakage tests (Section 7) monitor.

Chapter 4

Technology Stack

Table 4.1: Technology stack and rationale.

Layer	Choice	Rationale
Quantum framework	PennyLane (primary)	Native autograd + PyTorch interface; hardware-agnostic; cleanest hybrid-gradient workflow.
Quantum secondary Simulators	Qiskit 1.x + qiskit-ibm-runtime <code>lightning.qubit</code> (CPU), <code>default.qubit</code> (debug)	Real IBM hardware runs, transpilation and noise studies. ~100× faster than pure-Python simulation; practical to ~25 qubits on CPU.
Hardware access	IBM Quantum Open Plan (free tier)	1–2 real QPU runs for credibility; everything else simulated.
Classical ML	PyTorch, scikit-learn, XGBoost/LightGBM	XGBoost is the honest, strong baseline; not logistic regression.
Signal/imaging	MONAI, torchvision	Medical-grade ECG/image encoders for the hybrid QNN path.
Tabular data	pandas, Pandera	Schema validation catches silent data corruption at ingestion.
Imbalance	imbalanced-learn	SMOTE inside CV folds only (enforced by tests).
Explainability	SHAP, Captum, custom circuit diagnostics	End-to-end attribution in the <i>original</i> feature space.
Tracking	MLflow	Parameters, metrics, artefacts, circuit configs logged per run.
Backend	FastAPI + Pydantic	Async; automatic OpenAPI documentation.
Jobs	Celery + Redis	Quantum training takes minutes; never block HTTP.
Storage	PostgreSQL + MinIO/S3	Run metadata; artefacts (models, plots, reports).
Frontend	Streamlit (fast path)	Full experiment flow usable by a non-technical judge.
Packaging	Docker + docker-compose	One-command startup on a clean machine.
Testing	pytest + hypothesis	Unit, integration, leakage-guard, circuit property tests.

Layer	Choice	Rationale
Docs	MkDocs Material + model cards	Reproducible by an outsider without help.
Python	3.11, pinned versions	PennyLane/Qiskit edge cases on 3.12/3.13; lockfile committed.

Chapter 5

Datasets

Tier	Dataset	Size	Use	Access
1 (must)	Wisconsin Breast Cancer	569×30	pipeline sanity check	<code>sklearn.datasets</code>
1 (must)	UCI Heart Disease (Cleveland)	303×13	second tabular task	open
1 (must)	UCI Parkinson's (voice)	195×22	small- n regime test	open
2 (should)	PTB-XL ECG	21,837 records	real scale; CNN→quantum head; only dataset with demographics for fairness audit	PhysioNet, open
2 (should)	Pima Indian Diabetes	768×8	early-detection framing	open
3 (stretch)	TCGA gene expression (BRCA/LUAD)	$\sim 1000 \times 20000$	true high-dimensional compression showcase	GDC portal
3 (stretch)	MIMIC-IV sepsis onset	$\sim 300\text{k}$ stays	only if timeline allows (credentialing 1–3 weeks)	credentialed

Table 5.1: Dataset tiers. At least one Tier-2 dataset is mandatory for the full build: a project using only 569-sample breast cancer data is indistinguishable from two hundred existing GitHub repositories.

Chapter 6

Quantum Models

6.1 Encodings (Stage 4)

Encoding	Qubits for d features	Depth	Notes
Angle	d	$O(1)$	one rotation per feature; hardware friendly
ZZFeatureMap / IQP	d	1–3 layers	standard QSVM feature map [1]
Amplitude	$\log_2 d$ (16 features $\rightarrow 4$)	$O(d)$	exponentially compact; deep preparation
Data re-uploading	1–few	$O(\text{layers})$	features re-injected between layers; most parameter-efficient

Table 6.1: Quantum encodings. Encoding choice affects expressivity [6]; all four are implemented in `qednet.encoding`.

6.2 Quantum Kernel SVM (QSVM)

Encode each patient as $|\varphi(x)\rangle$ and compute the kernel matrix

$$K(x, y) = |\langle \varphi(x) | \varphi(y) \rangle|^2,$$

feeding K into `sklearn.svm.SVC(kernel="precomputed")`. **Cost wall:** the training kernel needs $\approx n(n-1)/2$ circuit evaluations; at $n = 1,000$ this is $\sim 5 \times 10^5$ circuits (feasible on a fast simulator), at $n = 5,000$ it is $\sim 1.25 \times 10^7$ (infeasible everywhere). This limit is reported as a *finding*, not hidden. An advanced variant, Quantum Kernel Alignment, makes the feature map itself trainable.

6.3 Variational Quantum Classifier (VQC)

Angle-encode $x \rightarrow L$ layers of strongly-entangling rotations $\rightarrow$ measure $\langle Z_0 \rangle \rightarrow$ sigmoid $\rightarrow$ binary cross-entropy. Training uses three gradient modes: backprop through the simulator (fastest), the parameter-shift rule (hardware-compatible, $2P$ circuits per step for P parameters), and SPSA (2 circuits per step, best under hardware noise). Gradient variance is logged every epoch as a barren-plateau diagnostic [2].

6.4 Data Re-uploading Classifier

Features are injected repeatedly between trainable single/few-qubit layers. Best parameter efficiency in the zoo — this is the model behind the “a 12-parameter quantum circuit matching a 400-parameter MLP” claim, measured honestly with a parameter-matched classical baseline.

6.5 Hybrid QNN

CNN/autoencoder backbone $\rightarrow$ bottleneck to 8 dims $\rightarrow$ VQC $\rightarrow$ dense head, trained end-to-end with autograd flowing *through* the quantum layer. Used on PTB-XL (21,837 records) where classical feature extraction is indispensable.

6.6 Classical Baselines

Logistic regression (L2, tuned C); Random Forest (tuned); **XGBoost with Optuna, 50+ trials** (the real opponent); RBF-SVM (direct counterpart of QSVM); and a **parameter-matched MLP** (the fair comparator for the efficiency claim). Hyperparameter budgets are equalised across models and logged — an unequal budget invalidates the entire comparison.

Chapter 7

Evaluation Methodology

7.1 Protocol

1. **Nested cross-validation:** outer 5-fold for performance, inner 3-fold for hyperparameters. No single 80/20 split is ever reported as a headline number.
2. **Multi-seed:** every experiment repeated over 10 random seeds; all results reported as mean $\pm$ standard deviation. A single-seed result is noise and is treated as such.
3. **Metrics:** AUROC, AUPRC, sensitivity at 95% specificity, F1, MCC, Brier score, ECE, decision-curve net benefit.
4. **Statistics:** DeLong test for paired AUROC comparison [11]; Nadeau–Bengio corrected paired t -test across folds [12]; bootstrap 95% CIs (2,000 resamples).
5. **Calibration:** reliability diagrams; Platt or isotonic correction fitted on a held-out calibration split.
6. **Cost axis:** wall-clock training time, inference latency, circuit count/depth, qubit count, parameter count, estimated QPU-seconds. Quantum loses badly on cost; we report it anyway — honesty is the credibility strategy.

7.2 Leakage Guard (automated)

A CI-enforced test suite asserts that every transformer (scaler, imputer, SMOTE, feature selector, PCA) is fitted exclusively on training folds — e.g., by checking that stored parameters equal train-fold statistics. A failing leakage test fails the build. This is rare in the field and is a headline engineering feature.

7.3 Expected Outcome (stated in advance)

Quantum models are expected to land within $\sim 1\text{--}3$ AUROC points of tuned XGBoost — sometimes marginally ahead on small datasets, usually behind — at $100\text{--}10,000\times$ the compute cost. That *is* the finding, and it is presented with a straight face.

Chapter 8

Explainability

1. **End-to-end SHAP:** the whole pipeline (raw features $\rightarrow$ prediction) is wrapped as one black box and KernelSHAP attributes importance in the *original* feature space [8]. Results are cached aggressively. (SHAP on the 8 post-PCA components is nearly useless clinically and is not presented as the primary output.)
2. **Linear back-propagation:** for PCA, attributions ϕ_z in compressed space map back as $\phi_x \approx W^\top \phi_z$.
3. **Quantum-native diagnostics:** input sensitivity $\partial\langle Z \rangle / \partial x_i$; per-layer entanglement entropy $S(\rho_A)$ (near-zero entropy means the circuit is just an expensive rotation); counterfactual explanations.
4. **Parameter-matched entanglement ablation** — the centrepiece experiment (below).

Parameter-matched ablation protocol. *Naive ablation (replacing entanglers with identity) confounds entanglement with model capacity, because it also removes parameters. QED-Net’s protocol: replace each two-qubit entangling gate with two trainable single-qubit rotations. Parameter count is held constant, so any performance change isolates entanglement itself. Run per-gate to produce an Entanglement Contribution Score ranking. Report the result whichever way it goes: a null result means the “quantum” model is an expensive parameterised rotation — which is itself the most valuable finding a team in this competition can present.*

Chapter 9

Novel Contributions and Features

9.1 N7: The Quantum Advantage Certificate

An automated referee that adversarially tests our own claims. Pipeline: (1) tuned classical baselines with equal, logged budgets; (2) parameter-matched MLP; (3) quantum models; (4) entanglement ablation; (5) *dequantization attack* — fit Random Fourier Feature and Nyström approximations of our quantum kernel; if they match performance, the kernel is classically simulable and there is no quantum advantage; (6) bottleneck honesty meter; (7) statistical tests; (8) cost normalisation (AUROC per QPU-second, per shot). Output: a signed one-page verdict — NO ADVANTAGE / PARAMETER-EFFICIENCY ADVANTAGE / SMALL-DATA ADVANTAGE / INCONCLUSIVE (UNDERPOWERED) — with evidence and a configuration hash. This contribution *cannot fail*: it produces a valid, publishable output whether quantum wins or loses.

9.2 F10: Bottleneck Honesty Meter

Train a simple linear classifier on the same 8 compressed features that the quantum model receives. The gap between that trivial model and the quantum model is what the quantum stage actually contributed. One day of work; pre-answers the question that destroys QML projects in Q&A.

9.3 F1: Quantum Second-Opinion Cascade

XGBoost screens every patient; only low-confidence cases ($p \in [0.35, 0.65]$, typically $\sim 8\%$) are routed to the slow quantum model. This converts quantum’s greatest weakness (speed) into a sensible deployment architecture.

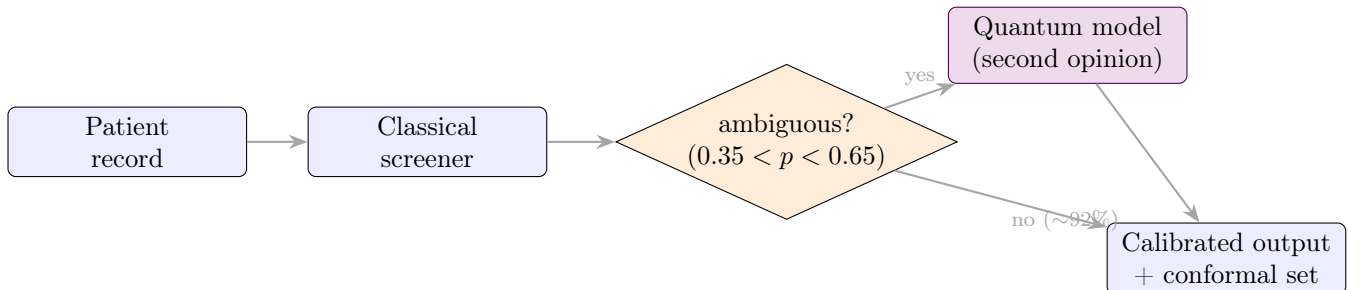


Figure 9.1: F1 cascade. Quantum cost is paid only on the ambiguous band; total system latency stays deployment-realistic.

9.4 F6 + N5: Conformal Abstention and Shot-Noise Uncertainty

Split-conformal prediction produces prediction sets with distribution-free coverage $\Pr[y \in C(x)] \geq 1 - \alpha$ [9]; a two-element set is an *abstention* — “escalate to a clinician.” Quantum measurement supplies a free, well-characterised aleatoric component: $\text{Var}(\hat{p}) = p(1 - p)/S$, combined with variance across circuit-initialisation ensembles for an epistemic component, and fed into the abstention rule. Caveat stated in the docs: shot noise is sampling uncertainty about a deterministic quantity, not model uncertainty; we test whether it correlates with real error and report either way. The abstention-rate/coverage trade-off curve is always reported.

9.5 Conditional and Deferred Features

- **F7 fairness audit** — only on PTB-XL (the sole dataset with sex/age); reports AUROC by subgroup and the quantum-vs-classical *fairness gap*.
- **F5 per-gate entanglement scores, F3 shot budgeting, F9 circuit cost compiler** — cheap extensions, added if time permits.
- **N2 pathway-entangled circuits and N6 federated QML** — only for a 4–5 person team; designs are documented in Future Work. (N2 caveat: STRING documents protein–protein interactions while TCGA provides mRNA expression; topology must be framed as hypothesis or built from co-expression networks.)

9.6 Deliverables Register

Table 9.1 lists all 20 deliverables with owners (A/B/C as defined in Chapter 10) and their status in solo mode (Chapter 11).

Table 9.1: Deliverables register with ownership and solo-mode status.

ID	Deliverable	Lead	Support	Est. hours	Solo-mode status
D1	Architecture & design document	All	—	12	Lite (in this doc)
D2	Data ingestion & validation	B	C	12	Lite (CSV only)
D3	Preprocessing & feature engineering	B	—	16	Full
D4	Quantum encoding library	A	—	10	Lite (angle + ZZ)
D5	Quantum model zoo	A	—	24	Lite (VQC + QSVM)
D6	Classical baseline suite	B	—	16	Full (XGBoost + MLP)
D7	Training/inference pipeline	C	A, B	12	Lite (CLI only)
D8	Evaluation & benchmarking engine	B	A, C	24	Lite (5-fold, 5 seeds)
D9	Explainability module	B	A	14	Lite (SHAP + ablation)

ID	Deliverable	Lead	Support	Est. hours	Solo-mode status
D10	Hardware execution report	A	—	8	Cut (optional)
D11	Backend API	C	—	12	Cut (Streamlit only)
D12	Web dashboard	C	—	20	Lite (single Streamlit page)
D13	Benchmark results report	A	B, C	16	Full
D14	Model cards	B	—	4	Full (2 cards)
D15	Deployment package	C	—	8	Lite (simple Dockerfile)
D16	Test suite & CI	C	B	10	Lite (leakage + unit tests)
D17	Documentation	C	All	8	Lite (README + this doc)
D18	Reproducibility bundle	C	B	4	Lite (seeds + lockfile)
D19	Ethics, privacy & regulatory note	B	All	4	Full (short)
D20	Demo video & pitch	C	All	8	Full (shorter)

Chapter 10

Team Plan: Three Members

10.1 Roles

Member A — Quantum Research Lead (“The Physicist”)

Owens everything inside the quantum core and the science writing.

- D4 encoding library; D5 model zoo (QSVM, VQC, re-uploading, hybrid QNN head)
- Barren-plateau diagnostics (gradient-variance logging); entanglement-entropy instrumentation
- Quantum half of the Advantage Certificate: ablation protocol, dequantization attack, circuit cost accounting
- D10 hardware run (transpilation, shot budget, error mitigation, noise study)
- Leads writing of D13 (benchmark report); co-authors D14/D19

Member B — Data & Evaluation Lead (“The Data Scientist”)

Owens the data spine and the evaluation rigour — the project’s credibility.

- D2 ingestion/validation; D3 preprocessing (leakage-safe by construction)
- D6 classical baselines with Optuna (equal budgets, logged)
- D8 evaluation engine: nested CV, all metrics, DeLong/bootstrap, calibration, DCA
- D9 explainability: end-to-end SHAP, honesty meter
- Dataset acquisition; PhysioNet credentialing *started in week 0*; model cards (D14); ethics note (D19) draft

Member C — Platform & Integration Lead (“The Engineer”)

Owens everything that makes it a product rather than a notebook.

- D7 config-driven training pipeline; package skeleton and the `fit/predict_proba` interface contract (week 1 — unblocks everyone)
- D11 FastAPI + Celery; D12 dashboard
- Integration of F1 cascade, F6 conformal, N5 uncertainty into the serving path
- D15 Docker; D16 CI and tests (with B’s leakage tests); D17 docs; D18 reproducibility; D20 demo video

10.2 Coordination Rules

1. **Week-1 interface freeze.** The package skeleton, config schema (Appendix A) and the estimator contract are frozen in week 1. After that, A, B and C code against interfaces, never against each other’s half-finished internals.
2. **Trunk-based development.** Short-lived branches, PRs reviewed within 24h, CI (tests + leakage guard) must pass to merge.
3. **Twice-weekly sync (30 min).** Monday: plan the week. Thursday: integration checkpoint against the current gate.
4. **Single source of truth.** Every experiment logged in MLflow; every headline number traceable to a run hash. No numbers in the report that MLflow cannot reproduce.
5. **Escalation path.** If any member is blocked >1 day on another’s module, they implement a mock behind the interface and proceed; the real implementation swaps in later.

10.3 Eight-Week Schedule

Table 10.1: Three-member schedule with weekly gates.

Week	Member A (quantum)	Member B (data/eval)	Member C (platform)	Gate
1	Environment; toy VQC on synthetic data	Datasets acquired; leakage-test framework; first XGBoost run	Repo, Docker, CI, package skeleton, interface freeze	docker compose up works; baseline v0
2	D4 encodings; QSVM end-to-end	D3 preprocessing; Optuna-tuned baselines (3 datasets)	D7 config-driven training; MLflow wiring	Quantum beats random on 2 datasets
3	D5 zoo complete; gradient diagnostics	Nested-CV engine; full metric suite	D11 API skeleton; D16 test suite	Auto-generated comparison matrix
4	Ablation protocol; certificate (quantum steps)	Statistics: DeLong, bootstrap, calibration	D12 dashboard v1	Certificate verdict on 2 datasets
5	Hardware prep: transpile, shot budget	End-to-end SHAP; honesty meter; model cards	Cascade + conformal in serving path	Mid-review; scope freeze
6	D10 hardware run; noise study	Fairness audit (PTB-XL, if in scope)	D15/D18 packaging; demo flow	All modules integrated
7	Leads D13 writing	D13 results tables; D19 ethics note	D17 docs; D20 video script	External reproduction test passes
8	Buffer; Q&A preparation	Buffer; polish statistics	Video edit; final packaging	Delivery

Chapter 11

Solo Mode: One Member Does Everything

This chapter is a deliberate contingency. Team-member attrition is the single most common cause of project failure; if two of three members become unavailable, the remaining member executes this plan unchanged and still delivers a complete, defensible project. The principle: **cut breadth, never cut rigour** — the leakage tests, multi-seed runs, ablation and certificate survive every cut, because they are the project’s grade.

11.1 Scope Reduction

Dimension	Three-member build	Solo build
Datasets	3–4 incl. PTB-XL	2 (breast cancer + heart disease)
Quantum models	4 (QSVM, VQC, re-uploading, hybrid QNN)	2 (VQC + QSVM)
Classical baselines	5, Optuna 50 trials	3 (XGBoost, MLP-matched, LogReg), Optuna 20 trials
Evaluation	Nested 5×3 CV, 10 seeds, full stats, DCA	Stratified 5-fold CV, 5 seeds, bootstrap CIs + DeLong, calibration
Certificate (N7)	Full (8 steps incl. cost normalisation)	Lite: baselines + ablation + dequantization + stats
Explainability	End-to-end SHAP + quantum diagnostics + counterfactuals	Pipeline-wrapped SHAP + parameter-matched ablation
Deployment	FastAPI + Celery + Postgres + dashboard	Single Streamlit app (train-from-cache, predict, explain, compare)
Hardware	1 real QPU run	Cut (a 2-qubit demo only if the queue permits)
Frontend	Streamlit full flow	Streamlit one-pager
Infrastructure	docker-compose stack	Single Dockerfile
Deliverables kept	D1–D20	D1, D3–D6, D8, D9-lite, D12-lite, D13, D14, D15-lite, D17-lite, D20

Table 11.1: Solo-mode scope. Rigour items (leakage tests, multi-seed, ablation, certificate-lite, calibration) are never cut.

11.2 Solo Schedule (8 weeks, 12–15 h/week $\approx$ 110–130 h total)

Table 11.2: Solo execution plan.

Week	Work	Output
1	Environment (Python 3.11, pinned); repo + simple CI; both datasets loaded; leakage-test framework; first XGBoost run	Baseline v0 table
2	Preprocessing module; 5-fold CV harness; metric suite (AUROC, AUPRC, sens@spec95, MCC, Brier)	Auto comparison table (classical)
3	VQC in PennyLane: angle encoding, 3-layer ansatz, Adam; gradient-variance logging	Quantum beats random on both datasets
4	QSVM + ZZFeatureMap; runtime measurements; both models into the harness	Full quantum-vs-classical matrix, 5 seeds
5	Parameter-matched ablation; certificate-lite (incl. RFF dequantization check); parameter-matched MLP baseline	Signed certificate verdict
6	Calibration (Platt); pipeline-wrapped SHAP; bottleneck honesty meter	Explanation plots in original features
7	Streamlit app (train-from-cache / predict / explain / compare); simple Dockerfile	Working demo
8	Report D13; two model cards; README; 6-minute demo video	Final delivery

11.3 Solo Kill Rules (decided now, not under panic)

- **Behind at end of week 3** $\Rightarrow$ drop the second dataset; concentrate on breast cancer.
- **Behind at end of week 5** $\Rightarrow$ drop QSVM (keep VQC); certificate-lite shrinks to ablation + stats.
- **Behind at end of week 6** $\Rightarrow$ feature freeze; spend remaining time on report, model cards and the video. A polished small project outscored a broken large one.
- **Never cut, under any circumstances:** leakage tests, multi-seed evaluation, the ablation experiment, and the honest reporting of negative results.

11.4 Degradation Path Summary

3 members: full build (Chapter 10). **2 members:** merge A and B into one science role; C keeps the platform but downgrades the dashboard to a Streamlit one-pager; drop D10 hardware and PTB-XL. **1 member:** solo build (this chapter). The evaluation spine and the certificate survive every mode.

Chapter 12

Risk Management

Table 12.1: Risk register.

Risk	Reality	Mitigation
Quantum loses on every metric	Likely	Headline claim is parameter efficiency / infrastructure, decided before results, not after
Simulator blowup	Statevector is $O(2^n)$; painful past ~ 24 qubits on CPU	Cap at 12–16 qubits; <code>lightning.qubit</code> ; document the ceiling
QSVM $O(n^2)$ cost	5,000 samples $\Rightarrow$ 12.5M circuits	Subsample for QSVM; report the wall as a finding
Barren plateaus	Training silently flatlines	Shallow circuits, local costs, layerwise training, live gradient-variance plots
Credentialing delays	PhysioNet 1–3 weeks	Start week 0; open-data fallback
Hardware queues	Free-tier queues can be hours	One late, small hardware run; simulator-only fallback
Data leakage	Kills credibility instantly	Automated leakage tests in CI
Regulatory exposure	“Detection” invites device questions	Research-only disclaimers in UI, docs, model cards
Demographic bias	Benchmarks are demographically narrow	Subgroup reporting where data permits; stated limitation
Reproducibility drift	Seed/backend nondeterminism	Pinned seeds, pinned versions, MLflow-logged everything
Team attrition	Common in student projects	Solo-mode plan (Chapter 11) is pre-agreed

Chapter 13

Ethics, Privacy and Regulatory Position

- **Status: research only.** The platform is explicitly labelled “not a medical device” in the UI, README and every model card. No clinical use, no patient-facing output.
- **Data protection.** Only open, de-identified benchmark data (or credentialed access under a signed DUA where applicable). Position with respect to HIPAA, GDPR and India’s DPDP Act is documented in D19; no PHI is processed by the platform itself, and the ingestion module strips direct identifiers.
- **Bias audit.** Subgroup performance reported where demographics exist (PTB-XL: sex, age); datasets’ demographic narrowness is stated as a limitation.
- **Transparency.** Negative results are published with the same prominence as positive ones; the Advantage Certificate exists specifically to make self-scepticism routine.

Chapter 14

Expected Outcomes and Success Criteria

Minimum publishable result (the floor, defined on day 1). *Even if every quantum model loses to XGBoost on every dataset, the project succeeds if it delivers: (i) the leak-proof, multi-seed harness; (ii) the Advantage Certificate’s signed verdict on at least two datasets with statistical tests; (iii) the parameter-matched ablation result, whichever way it goes; and (iv) an honest written report including negative results. This floor is set in writing so that no mid-project panic can lower the rigour bar.*

Stretch successes, in order of likelihood: parameter-efficiency advantage demonstrated (12-parameter circuit $\approx$ 400-parameter MLP); small-data robustness at $n = 50$ –200; performance parity with XGBoost within confidence intervals on at least one dataset; a non-trivial ablation result (entanglement contributing measurably to accuracy).

Chapter 15

Future Work

1. **N1 — Quantum survival classifier.** “Early detection” is fundamentally time-to-event. Discretise follow-up into K intervals; K measurement observables on a shared entangled state output conditional hazards h_k ; loss = discrete-time survival NLL; metrics = C-index, time-dependent AUC, integrated Brier score. Baselines: DeepSurv, Random Survival Forest. The largest unexplored gap in QML-for-health.
2. **N3 — Mixed-state missingness encoding.** Encode a missing value as $\rho = I/2$ (genuinely zero information) instead of imputing; partially-known values as partially mixed states. Requires $O(4^n)$ density-matrix simulation (≤ 8 – 10 qubits); the ancilla-flag variant runs on hardware.
3. **N4 — Cross-modal entanglement fusion.** Separate qubit registers per modality (imaging, genomics, clinical) fused by trainable cross-register entanglement; inter-register entanglement entropy doubles as a measured cross-modal dependency score.
4. **N6 — Federated QML.** Hospitals share only circuit parameters (~ 50 floats) via FedAvg — a communication-size privacy argument; differential-privacy noise required for any real privacy claim; gradient-inversion attacks have almost nothing to invert.
5. **F8 — Hardware drift suite.** Weekly QPU runs of a fixed golden set to track prediction drift; frames QML as a software-as-medical-device validation problem.

Chapter 16

Conclusion

QED-Net is a hybrid quantum–classical platform built on an unusual premise for this field: that the most valuable output is not a claim of quantum superiority, but a *machine that tests such claims adversarially*. The physics will not hand us an accuracy win over tuned XGBoost; our contribution is to be the team that can prove — with nested cross-validation, multi-seed statistics, leakage guards, a dequantization attack and a parameter-matched ablation — exactly what the quantum components do and do not contribute, and to wrap the surviving models in calibration, uncertainty and abstention so that they behave like honest research tools rather than demos.

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Appendix A

Example Experiment Configuration

One YAML file fully specifies a reproducible experiment; the hash of this file is embedded in every Advantage Certificate.

```
dataset:
  name: breast_cancer
  compression: pca
  target_dim: 8
model:
  type: vqc                # vqc | qsvm | reupload | hybrid_qnn
  n_qubits: 8
  layers: 3
  ansatz: strongly_entangling
  encoding: angle          # angle | zz | amplitude | reupload
  optimizer: adam
  learning_rate: 0.01
training:
  outer_folds: 5
  inner_folds: 3
  seeds: 10
  shots: 1024
evaluation:
  metrics: [auroc, auprc, sens_at_spec95, mcc, brier, ece]
  bootstrap_ci: 2000
  delong_test: true
  decision_curve: true
certificate:
  dequantization_attack: [rff, nystrom]
  entanglement_ablation: parameter_matched
  bottleneck_honesty_meter: true
```